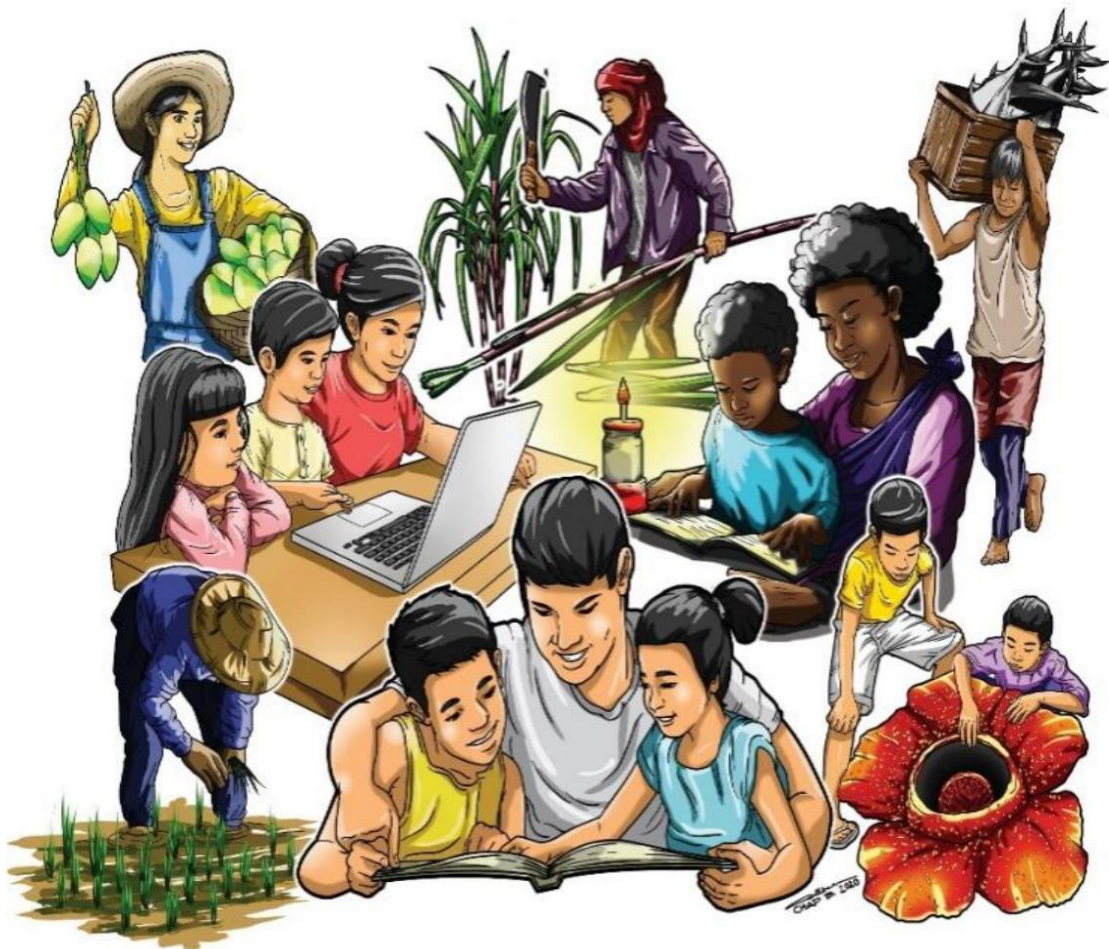


Science Activity Sheet

Quarter 3 – MELC 1

Week 1

Friction



REGION VI – WESTERN VISAYAS

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Science 6
Activity Sheet No. 1- Friction
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Introductory Message

Welcome to Science 6!

The **Learning Activity Sheet** is a product of the collaborative efforts of the Schools Division of Bacolod City and DepEd Regional Office VI - Western Visayas through the Curriculum and Learning Management Division (CLMD). This is developed to guide the learning facilitators (teachers, parents and responsible adults) in helping the learners meet the standards set by the K to 12 Basic Education Curriculum.

The **Learning Activity Sheet** is self-directed instructional materials aimed to guide the learners in accomplishing activities at their own pace and time using the contextualized resources in the community. This will also assist the learners in acquiring the lifelong learning skills, knowledge and attitudes for productivity and employment.

For learning facilitator:

The **Science 6 Activity Sheet** will help you facilitate the leaching-learning activities specified in each Most Essential Learning Competency (MELC) with minimal or no face-to-face encounter between you and learner. This will be made available to the learners with the references/links to ease the independent learning.

For the learner:

The **Science 6 Activity Sheet** is developed to help you continue learning even if you are not in school. This learning material provides you with meaningful and engaging activities for independent learning. Being an active learner, carefully read and understand the instructions then perform the activities and answer the assessments. This will be returned to your facilitator on the agreed schedule.

Name of Learner: _____
Grade and Section: _____ Date: _____

SCIENCE 6 ACTIVITY SHEET NO. 1

Friction

I. Learning Competency with Code

Infer how friction and gravity affect the movements of different objects.
(S6FE-IIIac-1)

II. Background Information for Learners

Friction plays a vital role in your everyday life. Whatever you do, wherever you go, friction is present. Even when you are sitting down, standing or holding a bottle or any object, friction affects your daily activities. Friction occurs between the surfaces of two objects rubbing or sliding against one another.

This learning activity helps you learn more about friction.

III. Activity Proper

A. Activity 1: Roll and Stop

Directions: You will need any round object available at home like a ball, a tin can, or a tomato and the floor. Write your answers on a separate sheet or in your science notebook.

Materials: ball or any round objects

Procedure

1. Get the ball or any round object and put it on the floor.
2. Push the ball or the round object gently to allow it to roll on the floor, as shown in Figure 1. You observe the movement of the ball.



Figure 1. A boy pushes a ball to roll on the floor

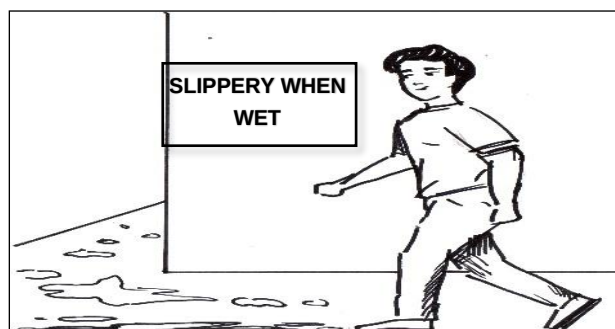
Guide Questions

1. What happened to the ball or the round object after it was pushed?

2. What happened to the ball or the round object after some time? Explain why the ball or the round object acted in such way.

Activity 2.

Directions. Study the picture and answer the questions.



Questions:

1. What do you think might happen if the boy stepped on the wet floor?

2. How can you prevent the possible accident from happening? Why?

What is friction?

Friction is a contact force that opposes or resists the change in the state motion of an object that is in contact with another object. This force is often called the *resistive force of friction*.

The resistance can result in *static friction*, where objects do not move with respect to another object. Or, it can be *kinetic or dynamic friction*, where the two objects in contact are moving. The static friction that holds an object in place is greater than the kinetic friction that slows down a moving object. In other words, once you start a car moving, the static friction holding the car in place *decreases* that converts to kinetic friction (Figure 2a and 2b).

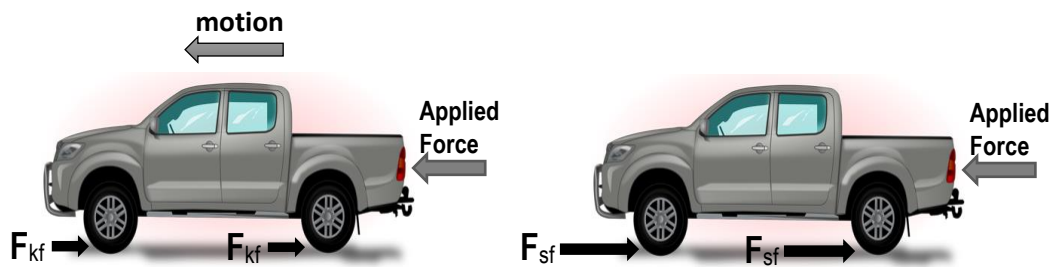


Figure 2. a) Kinetic friction; b) Static friction

Friction affects the motion of objects. It slows down or stops moving objects, but for objects at rest (not moving), it prevents them to move or slide (Figure 3). Friction always acts *opposite* the direction of an object's motion. Friction occurs between the surfaces of two objects rubbing against each other.



Figure 3. Friction a) slows down moving object; b) prevent the cabinet to slide or move

Table 1 below summarizes the types and uses of friction.

Table 1. Nature, types, and uses of friction

FRICTION		
Types	Description	Uses
* sliding friction	the resistive force that prevents two dry, solid objects in contact to slide.	* produces heat of fire, when the two surfaces rub each other * prevents the object from slipping or sliding
* rolling friction	the resistive force that slows the wheel's motion on the other solid surface	* slow down or stop cars * enables you to hold or grip things * keeps objects in place
* fluid friction-	the resistive force that acts on a solid object that is in contact with a fluid, such as a liquid or gas, the force is applied to either the object or to the fluid	* causes power loss * produces noise during any kind of operation * can wear and tear machine parts

Friction is considered undesirable, yet it plays an important role in your daily activities. Table 2 below tells the advantages and disadvantages of having friction and how to increase or reduce friction

Table 2. Advantages and Disadvantages of Friction and Ways to Increase or Reduce Friction

Advantages of Friction	Disadvantage of Friction
- It helps us walk and run on the ground - Brakes in a car make use of friction to stop the car - Asteroids are burnt in the atmosphere before reaching Earth due to friction. - It helps in the generation of heat when we rub our hands. - It helps to hold or grip on something or objects. - It enables us to write on paper using pens or pencils. - It keeps appliances, furniture, etc. be fixed in their place. - It helps athletes play their favorite sports.	- It produces unnecessary heat leading to the wastage of energy. - It slows down the motion of moving objects. - Forest fires are caused due to the friction between tree branches. - A lot of money goes into preventing friction and the usual wear and tear caused by it by using techniques like greasing and oiling.
	Ways of Reducing Friction
	- Use of lubricants in machines - Polishing the uneven surfaces - Use of ball bearings in rotating machines and shafts - Streamlining the bodies of cars or airplanes, boats, and ships to lessen fluid friction.
Ways of increasing Friction	
- Making the surfaces rough - Increasing the mass of the object that is moving.	

IV. Reflection

Is friction helpful or not? Why?

V. Answer Key

Activity 1: Roll and Stop

1. After pushing the ball, it rolled on the floor. Then it gradually decreased its motion or slowed down and eventually stopped.
2. The ball rolls on the floor when force is applied on it. The force of friction acts between the surface of the ball and the surface of the floor that result to an opposing or resisting force. The friction or opposing force between the two surfaces caused the ball to slow down and stop eventually.
3. Friction affects the motion of moving objects. It acts opposite the direction of motion of objects.

Practice Exercise:

1. The boy might slip because the water on the floor reduced friction acting between the floor and the shoes of the boy that would prevent him from slipping.
2. Mopping the floor to make it dry will prevent the possible accident to happen. Removing the presence of water will create a greater friction that will help prevent the boy from slipping.

Reflection:

Friction is helpful because it helps prevent accidents by increasing friction on roads and floors. It is also helpful in producing heat. Friction keeps non-moving objects in place like when we are sitting on a chair.

VI. Other References

Book

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